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Isolation of three new meroterpenoids and seven known compounds from *Albatrellus yasudae* and their Aβ-aggregation inhibitory activity

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ABSTRACT

Alzheimer's disease is a serious neurologic disorder that cannot be cured completely. In this study, we targeted compounds that inhibit amyloid-beta (Aβ) aggregation, based on the amyloid cascade hypothesis. Ten compounds (**1**-**10**) were isolated from $CHCl_3$ extracts of the mushroom *Albatrellus yasudae* using Aβ-aggregation inhibitory activity–guided separation. The structures of these compounds were elucidated from 1D and 2D NMR and MS spectral data. Compounds **1**-**3** were novel, whereas **4**-**10** were identified as the known compounds grifolin, grifolic acid, neogrifolin, confluentin, 2-hydroxyneogrifolin, daurichromenic acid, and a cerebroside derivative. Compounds **1**-**10** were tested for Aβ-aggregation inhibitory activity. Compounds **1**, **3**, **5**, **6**, **8**, and **9** have potential as Aβ-aggregation inhibitory activity.

Alzheimer's disease (AD) accounts for 60% of dementia cases, and its incidence is increasing. Several drugs are available to treat AD based on two different hypotheses: acetylcholinesterase inhibitors (e.g., donepezil, rivastigmine, galantamine), and NMDA receptor antagonists (e.g., memantine). However, these drugs treat symptoms and cannot completely cure AD. Our study was based on the amyloid cascade hypothesis.[1] This hypothesis holds that amyloid-beta (Aβ) precursor protein (APP) present on neuronal cell membranes is cleaved by β-secretase (BACE1) and γ-secretase, generating Aβ peptides. Aβ subsequently aggregates and is deposited in neurons, ultimately leading to AD. According to this hypothesis, we targeted inhibitors of Aβ aggregation.

Albatrellus yasudae (Lloyd) Pouzar is a Basidiomycota-Agaricomycetes-Russulales-Albatrellaceae mushroom. The characteristic fruiting body of *A. yasudae* is shiny and dark blue indigo in color. The fruiting body is also edible. Various useful compounds have been isolated from other members of the same family as *A. yasudae*.[2-4]

Our laboratory has collected numerous extracts of various mushrooms, and these were examined for possible inhibitors of Aβ aggregation. A $CHCl_3$ extract of *A. yasudae* exhibited inhibitory activity of Aβ aggregation, and we therefore sought to identify the responsible compound(s).

Fruiting bodies of *A. yasudae* were collected in Yamanashi Prefecture, Japan (2015, September). A voucher specimen was deposited at our laboratory (SYC-53). A $CHCl_3$ extract of dried fruiting bodies of *A. yasudae* was successively subjected to chromatography over a silica gel column, ODS column, preparative HPLC, preparative TLC, Sephadex LH-20, and recrystallization. Separated by bioassay-guided fractionation, yielding ten compounds (**1**-**10**). The structures of three novel compounds (**1**-**3**), were determined as 2-hydroxy-1-methoxy neogrifolin (**1**), 9'-keto-grifolic acid (**2**), and bis-2-hydroxy-1-methoxy neogrifolin (**3**) (Table 1). Seven known compounds were also identified as grifolin (**4**),[2] grifolic acid (**5**),[5] neogrifolin (**6**),[6] confluentin (**7**),[7] 2-hydroxyneogri-folin (**8**),[8] daurichromenic acid (**9**),[9] and cerebroside derivative, ((*R*)-2-hydroxy-*N*-((2*S*,3*R*,4*E*,8*E*)-3-hydroxy-9-methyl-1-(((2R,3R,4S,5S,6R)-3,4,5-trihydroxy-6-(hydroxymethyl)tetrahydro-2H-pyran-2-yl)oxy)octadeca-4,8-dien-2-yl)heptadecanamide) (**10**) [10] (Figure 1). The structures of compounds **4**-**10** were determined based on their NMR spectra.

Compound **1** was obtained as a colorless oil, and its molecular formula was established as $C_{23}H_{34}O_3$ by HREIMS (*m/z* 358.2509 $[M]^+$, calcd. 358.2508). The IR spectrum indicated absorption of hydroxy groups at 3404 cm^{-1}. ^{1}H- and ^{13}C-NMR, DEPT, and HMQC data for **1** indicated six methyls, five methylenes, four methines, and eight quaternary carbons. The ^{1}H-NMR spectrum revealed the presence of one aromatic proton at δ_H 6.55 (s, H-4), five methyls (at δ_H 1.67, 1.57, 1.59, 1.76, and 2.16, due to H-12'-15' and H-7), one methoxy group at δ_H 3.77 (s, H-8), and three olefinic protons at δ_H 5.05 (3H, t) due to H-2', H-6', and H-10'. The ^{1}H- and ^{13}C-NMR spectra of **1** were similar to those of neogrifolin (**6**), except that the C-1 hydroxy group was substituted with a methoxy group, and the C-2 olefinic proton was substituted with a hydroxy group. The positions of the methoxy and hydroxy groups of **1** were determined based on HMBC correlations, as shown Figure 2. Thus, the structure of compound **1** was established as 2-hydroxy-1-methoxy neogrifolin.

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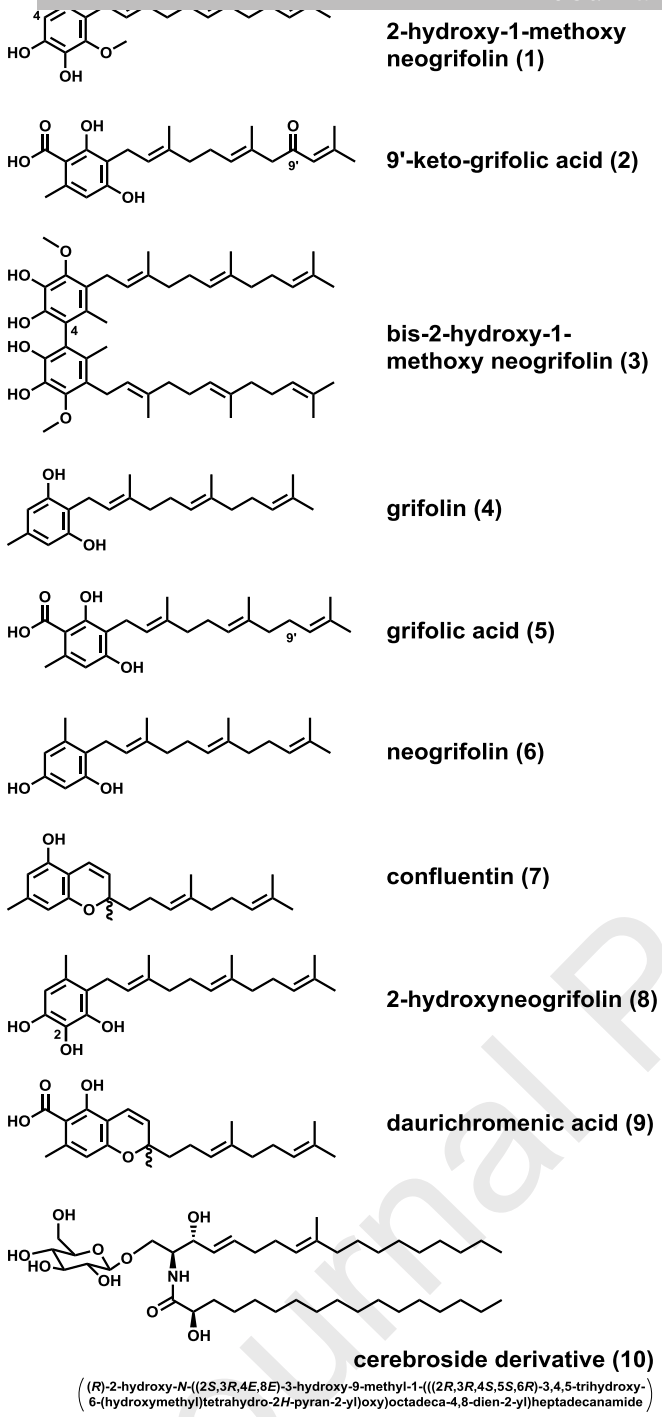

Figure 1. Isolated compounds (**1-10**).

Compound **2** was obtained as a pale-yellow solid that gave an $[M]^+$ ion in HREIMS at *m/z* 386.2097 (calcd. for $C_{23}H_{30}O_5$, 386.2093), consistent with the molecular formula $C_{23}H_{30}O_5$. The IR spectrum of **2** revealed absorption at 3182 & 1666 cm^{-1} (COOH), 2970, 2930 and 2856 cm^{-1} (CH), and 1619, 1605 and 1455 cm^{-1} (C=C). The ^{1}H- and ^{13}C-NMR, DEPT, and HMQC data for **2** indicated five methyls, four methylenes, four methines, and ten quaternary carbons. The NMR spectrum of **2** was very similar to that of grifolic acid (**5**). Compound **2** had one more carbonyl group and one less methylene than grifolic acid, and the HMBC correlations from H-8' (δ_H 2.93) and H-10' (δ_H 6.13) to C-9' (δ_C 202.2) revealed that the carbonyl group is located at position 9 (Figure 2). Thus, the structure of compound **2** was established as 9'-keto-grifolic acid.

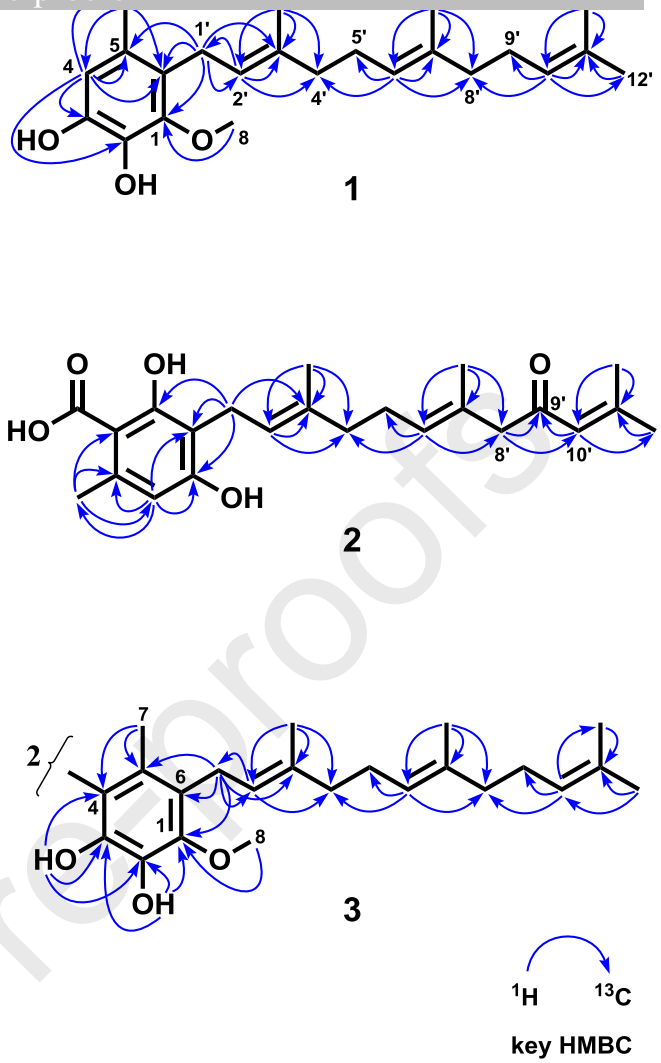

Figure 2. Key HMBC correlations of **1-3**.

Compound **3** was obtained as a yellow oil that gave an $[M]^+$ ion in HRFABMS at *m/z* 714.4850 (calcd. for $C_{46}H_{66}O_6$, 714.4859), consistent with the molecular formula $C_{46}H_{66}O_6$. However, the ^{1}H- and ^{13}C-NMR spectra of **3** were very similar to those of **1**. Thus, compound **3** was deemed to be a dimer of **1**. The binding site of the dimer was determined based on disappearance of the H-4 proton in the ^{1}H-NMR spectrum of **3** and HMBC correlations from H-7 (δ_H 1.84) and OH-3 (δ_H 4.79) to quaternary C-4 (δ_C 117.4) (Figure 2).

The ten isolated compounds (**1-10**) as well as the methyl ester of **5** (**5a**)[11] were examined for Aβ_{40}-aggregation inhibitory activity using the Th-T assay.[12,13] The IC_{50} value of myricetin (positive control) and compounds **1-10** and **5a** was shown in Table 2. Compounds **1**, **3**, **5**, **6**, **8**, and **9** exhibited Aβ-aggregation inhibitory activity. Especially, compound **3** showed the similar strong activities with positive control, myricetin. Compound **5** exhibited inhibitory activity, but the methyl ester of **5** (methyl grifolate **5a**) did not inhibit Aβ_{40} aggregation. Compound **7** and **9** (having pyran ring that formed by the part of farnesyl group as side chain) decreased Aβ-aggregation inhibitory activity too. These data support the hypothesis that carboxyl and farnesyl groups play an important role in the inhibition of Aβ_{40} aggregation. Compound **2**, in which the farnesyl group of grifolic acid is oxidized, exhibited no inhibitory activity, indicating that a normal farnesyl group is required for inhibitory activity.

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Table 1 ^{1}H- and ^{13}C-NMR Data for Compound **1-3**

	1		**2**		**3**	
	δ ^{1}H	δ ^{13}C	δ ^{1}H	δ ^{13}C	δ ^{1}H	δ ^{13}C
1		145.6		141.9		145.8
2		142.0		105.1		139.9
3		134.0	6.19 (d, 0.5)	111.4		134.9
4	6.55 (s)	113.0		161.1		117.4
5		124.8		113.9		127.8
6		128.9		164.8		126.3
7	2.16 (s)	19.0		175.6	1.84 (s)	15.4
8	3.77 (s)	61.5	2.46 (s)	24.3	3.87 (s)	61.1
1'	3.28 (d, 6.4)	25.7	3.27 (d, 7.3)	22.7	3.36 (d, 6.0)	25.8
2'	5.05 (t, 6.4)	123.0	5.24 (dd, 7.3, 1.3)	124.7	5.08 (m)	124.4
3'		135.0		134.6		135.3
4'	2.00 (m)	39.6	2.00 (t, 7.4)	40.5	2.00 (m)	39.7
5'	1.93 (m)	26.5	2.12 (quin, 7.4)	27.5	2.00 (m)	26.8
6'	5.05 (t, 6.4)	124.0	5.21 (dd, 7.4, 1.3)	130.6	5.08 (m)	124.1
7'		131.3		130.3		135.1
8'	2.01 (m)	39.7	2.93 (s)	56.2	2.00 (m)	39.7
9'	2.01 (m)	26.7		202.2	2.00 (m)	26.6
10'	5.05 (t, 6.4)	124.3	6.13 (quin, 1.3)	123.8	5.08 (m)	123.1
11'		135.3		157.7		131.3
12'	1.67 (s)	25.7	1.85 (d, 1.3)	27.7	1.67 (s)	25.7
13'	1.57 (s)	17.7	2.09 (d, 1.3)	20.8	1.57 (s)	17.7
14'	1.59 (s)	16.2	1.53 (d, 1.3)	16.3	1.57 (s)	16.0
15'	1.76 (s)	16.0	1.76 (d, 1.3)	16.1	1.76 (s)	16.3
	5.23 (OH)				5.31 (OH)	
	5.07 (OH)				4.79 (OH)	

Table 2

Aβ$_{40}$-aggregation inhibitory activity of compounds **1-10** and **5a**

compound	IC_{50} (μM)
1	29.3
2	150.8
3	12.3
4	148.8
5	20.3
6	19.9
7	68.0
8	16.7
9	24.7
10	62.7
5a	221.0
myricetin	7.4

*Each test was performed in duplicate

In conclusion, we isolated ten compounds from *A. yasudae*. Compounds **1-3** were novel, whereas **4-10** were known compounds. And evaluated their Aβ$_{40}$-aggregation inhibitory activity, it was revealed carboxyl and farnesyl groups play an important role in the inhibition of Aβ$_{40}$ aggregation.

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Appendix A. Supplementary data

The structures of compounds **4-10** were identified by comparing their physical and spectral data with reported values[2, 5-10]; see Tables S1, 2.

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11. Methylation of grifolic acid; Grifolic acid (10 mg, 0.027 mmol) was dissolved in MeOH (1 mL), stirred for a few minutes, and then TMS-diazomethane (0.6 mL, 0.36 mmol, 13.4 eq) was added to the reaction solution in a dropwise manner on ice. After stirring overnight at room temperature, the reaction was quenched by adding an excess amount of MeOH, after which the resulting residue was purified by Si. C. C. (10 g, $CHCl_3$-MeOH). Target material (8.5 mg; 82.4%, colorless oil) was then obtained.
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Supplementary material

Supplementary data associated with this article can be found in the online version, at

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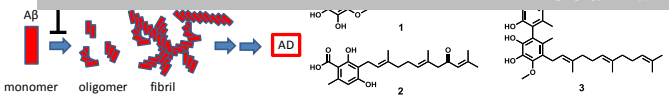

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